Dynamic NBA Team Ratings and Pregame Win Probabilities

Model Specification, Calibration, and Historical Evaluation

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Abstract

An NBA forecasting system maintains team ratings and estimates pregame win probabilities using a dynamic multifactor rating model. Rating differences, home court, rest, estimated travel, and time-zone changes determine pregame probabilities. Outcomes, schedule strength, lead retention, and box-score performance determine subsequent rating updates. Model version 0.2.0 is fitted on 2021–22 through 2023–24 regular-season history and frozen before evaluation on 2024–25 and 2025–26. Five validated full seasons contain 6,150 games. On 1,075 eligible games in the held-out 2025–26 season, log loss is 0.5925, Brier score is 0.2035, and winner accuracy is 68.56%. These are reconstructed historical forecasts, not predictions issued live. The manuscript specifies feature derivations, bounded parameter estimation, data repairs, and the limitations of this first NBA calibration.

1 Scope

The system produces teams ordered by rating, matchup probabilities, and a record of forecasts and outcomes. Live ratings update after completed games; in-game probability is outside scope. The model uses explicit feature units, stable transformations, one complementary probability function, and NBA-specific calibration. The following sections specify the equations, data construction, fitting protocol, and evaluation. The website, historical replay, and manually invoked live collector are implemented. Automatic collection is scheduled through GitHub Actions; frontend hosting is deployed with public access.

2 Forecast timing

Let τ_g be a forecast cutoff and $\mathcal{F}_{\tau_g}$ the information available by then. A home-team forecast estimates $p_g = \Pr(y_g = 1 \mid \mathcal{F}_{\tau_g})$, where y_g is the home-win indicator. The historical replay freezes all forecasts for an Eastern-calendar daily slate at the earlier of 10:00 America/New_York and one hour before its first tip-off. It requires all prior slates to have completed before that cutoff; current-slate results cannot enter its forecasts. Ratings update after the slate. A team cannot appear twice on one slate, and duplicate games are rejected.

Completion availability is proxied by terminal EndGame wallclock timestamps, rather than later edited play timestamps. Seventeen games use dated final-recap publication times. One game with source play dates shifted by a day uses an explicitly flagged conservative tip-off-plus-eight-hours

availability bound; this is not an observed completion time. Replays check these proxies against the next cutoff but cannot establish when historical corrections first reached a live feed. Genuinely issued forecasts record their actual issue time, data cutoff, and model version in a separately preserved ledger.

3 Pregame probability

For home team i and away team j, define pregame ratings r_i, r_j, rest RD in days, travel TD in kilometres, and time-zone change TZ in hours. Let $h = 1$ at a home venue and $h = 0$ at a neutral venue. The fatigue indices, with an empirically fitted travel coefficient ν, are

$$F_i = RD_j + \exp(|TZ_i|/3) + \nu TD_i, \tag{1}$$

$$G_i = RD_i + \exp(|TZ_j|/3) + \nu TD_j. \tag{2}$$

They are empirical indices, not physiological measurements. The probability is

$$p_i = [1 + 10^{z_i}]^{-1}, \qquad z_i = \frac{r_j - r_i - Hh}{c} + \gamma \ln(F_i/G_i), \qquad p_j = 1 - p_i. \tag{3}$$

All logarithms are natural unless explicitly subscripted. Equivalently,

$$\operatorname{logit}(p_i) = \ln(10) \left[\frac{r_i - r_j + Hh}{c} - \gamma \ln(F_i/G_i) \right]. \tag{4}$$

The exponent base and rating scale cannot both be freely identified. Moreover, rescaling ratings, K, H, and c together preserves probabilities. We fix the base at 10 and $c = 993.39 \ln(10)/\ln(99.39) = 497.3558141396255$ to anchor the rating scale, then fit K, H, and γ. Display rounding does not affect calculations.

4 Rating updates

For either team, let y denote its own win indicator, p its complementary pregame probability, and $s = 2y - 1$. The implemented update is

$$r^+ = r + K(y - p)(1 + s\beta S)(1 + \phi \log_2(1 + L))(1 + s\lambda(Q - 1)). \tag{5}$$

Here β is a fitted schedule-strength multiplier, ϕ weights lead retention, and λ weights relative performance. Both teams use their pregame ratings and their own performance history. The result and all current-game box-score features enter only after completion. Each season starts independently at 1500. Asymmetric updates can shift the league mean; recentering it at 1500 after each slate preserves all pairwise differences. Inter-season carryover is not part of this release.

4.1 Schedule strength

For team i facing j, define

$$S_i = \frac{2W_j + \overline{W}_{\mathcal{O}(j)}}{3}, \qquad W_t = \frac{w_t + 1}{n_t + 2}. \tag{6}$$

All wins w_t and games n_t precede the cutoff. The second term averages smoothed win rates of j's previous opponents, including repeated opponents in proportion to their frequency. Empty opponent history receives 0.5. This schedule-strength feature uses only results available before the cutoff.

4.2 Lead retention and relative performance

For largest leads LL_i, LL_j and final scores PTS_i, PTS_j, use

$$L = \frac{|PTS_i - PTS_j|}{\max(LL_i, LL_j)}. \tag{7}$$

Using the larger of both peak leads defines the ratio even for teams that never led. Valid non-tied completed games satisfy $0 < L \leq 1$; peak leads below the final margin are rejected. This measures final margin relative to either side's peak advantage, not a separate losing-team retention ratio.

For the eight Appendix A features, fixed scales are

$$\widetilde{\boldsymbol{X}} = (X_1/10, X_2, X_3, X_4, X_5, X_6/3, -X_7, X_8). \tag{8}$$

The turnover sign rewards fewer turnovers. Positive weights sum to one:

$$U = \sum_{k=1}^{8} \alpha_k \widetilde{X}_k, \tag{9}$$

$$A = \text{clip}\left((1 + e^{-U})^{-1}, 10^{-6}, 1 - 10^{-6}\right), \tag{10}$$

$$Q = \text{clip}\left(A/\overline{A}_{<g}, 0.25, 4\right). \tag{11}$$

A is the aggregate game performance factor (AGPF), and Q is the performance adjustment factor (PAF). The reference is the expanding mean of that team's prior games only. Openers use $Q = 1$. The sigmoid maps the aggregate to a bounded positive value while preserving signed net efficiency in its input. With $0 \leq \lambda \leq 0.25$ and the stated PAF range, the performance multiplier cannot become negative. These transformations and scales were fixed before fitting; the eight weights were fitted.

5 Data construction

Free ESPN-derived schedule, team-box-score, and play-by-play parquet files from SportsDataverse provide the initial history [1]. Primary ESPN game summaries replace incomplete or inconsistent archival records [2]. GeoNames supplies venue-city coordinates and IANA time zones [3]. Each of the five seasons contains 1,230 completed standard regular-season games, 30 teams, and 82 games per team. Preseason, playoffs, All-Star games, and the NBA Cup final are excluded; Cup games that count toward the regular season remain included.

Validation requires unique game/team identities, matching opponent rows and header scores, finite nonnegative integer counts, made shots no greater than attempts, scoring identities, plausible assists and steals, valid feature denominators, and peak leads consistent with final margins. Missing values are not converted to zero. Largest leads can be recovered from a score path only when its final

scores match the header. Inconsistent team aggregates can be replaced by complete player rows with matching points and shooting identities. Turnovers consistently mean summed player turnovers, excluding provider non-player team turnovers. Player and provider non-player team turnovers are kept distinct.

Season	Accepted games	Primary replacements	Unresolved failures
2021–22	1230	132	0
2022–23	1230	158	0
2023–24	1230	201	0
2024–25	1230	165	0
2025–26	1230	217	0

Table 1: Source audit. The 873 primary replacements mostly address incomplete archived play-by-play; they are not a count of corrupt final box scores.

All required numerical formula inputs are complete. Repair actions, availability proxies, source URLs, retrieval times where available, and SHA-256 hashes are stored with the data or collection manifests. This establishes internal consistency, not perfect measurement of every real-world quantity. Travel is estimated between successive venue cities, not actual itineraries. Rest counts intervening Eastern dates, so back-to-backs have zero. Openers receive three rest days and zero travel/time-zone change, with an initialization flag. Both venue offsets are evaluated at the current tip-off instant, preventing a daylight-saving change alone from creating geographic travel. Ambiguous or unknown cities cause validation failure.

6 Parameter estimation

Training uses 2021–22 through 2023–24; 2024–25 is validation and 2025–26 is a held-out test. The objective is mean binary log loss,

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{g}[y_g \ln p_g(\theta) + (1-y_g)\ln(1-p_g(\theta))]. \tag{12}$$

A game is scored after both teams have played ten prior games, giving 3,220 training observations. Earlier games still initialize histories, update ratings, and retain forecasts. Every candidate recomputes parameter-dependent ratings and AGPF history. An accelerated Numba replay is checked against the canonical Python engine before fitting and at the fitted solution, to probability tolerance 10^{-11}.

SciPy differential evolution with L-BFGS-B polishing [4] fits 14 free dimensions: seven scalar coefficients and seven softmax logits. With the eighth logit fixed to zero, $\alpha_k = \exp(u_k)/\sum_l \exp(u_l)$ and $u_8 = 0$. Travel is optimized on a base-10 logarithmic scale. Settings are random seed 42, population multiplier 8, maximum 150 generations, relative tolerance 10^{-5}, one worker, and polishing enabled. The run converged in 44 generations and 5,520 objective evaluations. Training log loss decreased from 0.637588 at the reference starting configuration to 0.628210. Constants were frozen at 03:04:25 UTC on October 13, 2026 (October 12 Eastern), before validation/test evaluation. No parameters or formula choices were changed after those scores.

Parameter	Search interval	Fitted value
K	$[1, 80]$	12.217347
H	$[0, 180]$	41.552847
γ	$[0, 0.5]$	0.164724
ν	$10^{[-8,-1]}$	0.000109735
ϕ	$[0, 2]$	2.000000
λ	$[0, 0.25]$	0.250000
β	$[0, 1]$	0.978548
$u_1, \ldots, u_7$	$[-5, 5]$	softmax weights below

Table 2: Rounded fitted constants. Full precision is retained in the versioned configuration. Base 10, c, initial rating 1500, and time-zone scale 3 are fixed.

α_1	α_2	α_3	α_4	α_5	α_6	α_7	α_8
0.161954	0.004365	0.006380	0.002774	0.686992	0.073041	0.007926	0.056568

The lead and performance coefficients reached their upper bounds. This is a converged bounded solution, not proof of a global optimum or uniquely identified causal effects. Fixed feature scales also influence weight interpretation. Future constraint changes require a new model version and new prospective evidence; the observed test season is no longer untouched for such revisions.

7 Historical evaluation

Probability quality is assessed by log loss, Brier score $N^{-1} \sum_g (p_g - y_g)^2$, and ten equal-width calibration bins with counts, mean forecasts, and observed home-win fractions. Lower log loss and Brier score are better. Winner accuracy is secondary, choosing home at exactly $p = 0.5$. It is not an accuracy percentage for probabilities.

Season	Role	Scored	Log loss	Brier	Winner accuracy
2021–22	Training	1073	0.6298	0.2191	66.64%
2022–23	Training	1073	0.6457	0.2273	62.26%
2023–24	Training	1074	0.6092	0.2109	66.01%
2024–25	Validation	1075	0.6043	0.2088	66.42%
2025–26	Held-out test	1075	0.5925	0.2035	68.56%

Table 3: Frozen-model chronological replay. All 1,230 forecasts per season are retained; primary scoring excludes games before both teams' ten-game warmup.

Across five seasons, 5,370 games are scored and 780 are warmup-only. Test ratings continue updating after observed outcomes, matching an adaptive rating system while coefficients remain frozen. Training scores are in-sample. A single held-out season does not establish performance across all future seasons, and sparse extreme-probability bins limit calibration inference. The recorded bin summaries are descriptive; no uncertainty-adjusted calibration or repeated-season confidence claim is made.

8 Simple Elo comparison and chronological diagnostics

A comparator estimates only K and H, with the same fixed c, season initialization, cutoff, and warmup as the full model. Its equations are

$$p_i = [1 + 10^{(r_j - r_i - Hh)/c}]^{-1}, \qquad r_i^+ = r_i + K(y_i - p_i), \quad r_j^+ = r_j - K(y_i - p_i). \tag{13}$$

Fatigue, schedule, lead, and performance coefficients are zero. Both models fit training log loss with identical bounds on K and H, seed 42, and the same differential-evolution and polishing settings. Comparator predictions are checked against the canonical engine. Data and scored games are paired. This diagnostic was added after viewing original results; it is not a new untouched test or prospective evaluation.

First, compare the original three-season frozen configuration with simple Elo fitted on those same seasons:

Season	Model	Scored	Log loss	Brier	Accuracy
2024–25	Multifactor	1075	0.6043	0.2088	66.42%
2024–25	Simple Elo	1075	0.6137	0.2129	65.95%
2025–26	Multifactor	1075	0.5925	0.2035	68.56%
2025–26	Simple Elo	1075	0.6018	0.2073	66.79%

Second, expanding-window checks fit the first available season and evaluate the next, then add one training season per fold. Only the training prefix enters either objective. Full-model equations and bounds remain fixed; the three-season full fit is reused from the original release. Other fits are research configurations and do not overwrite archived release predictions.

Evaluation	Train seasons	Scored	Full LL	Simple LL	Full BS	Simple BS
2022–23	1	1073	0.6469	0.6522	0.2280	0.2300
2023–24	2	1074	0.6105	0.6168	0.2114	0.2145
2024–25	3	1075	0.6043	0.6137	0.2088	0.2129
2025–26	4	1075	0.5920	0.6013	0.2033	0.2071

All added optimizations converged. The full model has lower log loss and Brier score in all four folds. Winner accuracy improves in three folds but falls from 62.63% to 61.32% in 2022–23. Probability-score improvement therefore does not imply improvement in every classification statistic.

Paired exploratory uncertainty estimates use 10,000 bootstrap resamples of whole Eastern-calendar daily slates. Each sampled slate retains its eligible games, and differences are weighted by the total sampled game count. This preserves within-slate dependence but does not model dependence across days or fitting uncertainty; intervals have no multiple-comparison adjustment. For the original frozen model on 2025–26, the log-loss difference (full minus simple) is -0.00928, with exploratory 95% interval $[-0.01417, -0.00395]$. The analogous expanding-window interval crosses zero in 2022–23 and is below zero in the other three folds. These intervals are descriptive evidence rather than a guarantee of future improvement.

Across the four full fits, K ranges from 11.134 to 13.085, H from 33.368 to 47.570, and γ from 0.1647 to 0.1829. Individual travel and performance weights are less stable: ν ranges from 1.32×10^{-7} to 1.10×10^{-4}, and the defensive-rebound weight α_5 ranges from 0.4795 to 0.9610. Lead and

performance coefficients remain at their upper bounds in every fit. The checks support a modest repeated probability-score advantage for the model family, while leaving uncertainty about unique feature effects and overfitting. Attributing improvement to individual features would require an ablation study.

The original v0.2 configuration is retained provisionally for live integration. The four-season refit is not promoted merely because its now-observed score is slightly better. Source hashes, per-fold fit reports, metrics, paired predictions, and intervals are saved under the comparison release directory. Future issued forecasts will provide prospective monitoring. The historical outcomes were already available during development, so these expanding-window checks must not be presented as newly untouched evidence.

9 Reproducibility and application

The repository stores clean game records, two-row-per-game box-score tables, derived-feature tables, and reconstructed forecasts under season directories. Pregame and postgame feature columns are explicitly distinguished. Model configuration, optimizer bounds, seed, convergence trace, coverage audits, and source manifests identify the release. Large raw parquet downloads and ESPN response caches stay outside Git. The documented scripts reproduce acquisition, validation, fitting, and export; provider archives may change, so hashes identify the exact source snapshots used here.

Streamlit displays Teams, Games, and Methodology for a shared selected season, defaulting to the current season. Archived seasons expose scores and downloadable CSV files. The first observed 2026–27 snapshot holds initial ratings and three issued upcoming regular-season forecasts. No regular-season results have arrived in that snapshot; preseason outcomes are excluded. Light/dark appearance modes and fixed dropdowns are supported. The manual collector validates completed games, applies the frozen model, and saves upcoming forecasts without fitting during page interactions. GitHub Actions schedules collection every two hours. Failed ingestion must preserve the last valid output and expose freshness. Corrected records may trigger rating replay but cannot overwrite genuinely issued forecasts. The repository and hosted application are publicly accessible.

10 Observed after-game pipeline

The current-season collector refreshes individual ESPN scoreboard dates and validated final summaries. Provider whitelist calendar metadata restricts requests to known season dates; the default checks a seven-day lookback, unresolved events, and seven Eastern-calendar dates ahead. A full-refresh option revisits the season calendar for older corrections. Unknown regular categories, missing known finals, incomplete box scores, or inconsistent identities block publication. Preseason, All-Star, and Cup-final events remain excluded. A single-writer filesystem lock and atomic bundle replacement protect concurrent updates and preserve a valid snapshot on ingestion failure. The application shows last-successful freshness and failed-attempt status.

Rating-state reconstruction and forecast issuance are separate. An internal timeline predicts the rating-update expectation at each tip-off from earlier source-completion events, then updates and recenters at individual completions. These estimates enable backfill and correction replay, but are not treated as genuinely issued predictions. Completion proxies do not establish when a live provider first made every historical statistic available.

Observed runs issue probabilities only for scheduled games still in the future, using an actual timestamp after source retrieval; the CLI cannot backdate. A change in rating state, configuration, or announced context creates a new revision. Identical inputs reuse the previous revision. Current ratings are held fixed while announced intervening schedules project future rest and venue travel; unknown future results are never fabricated. Revisions retain their probabilities, timestamps, input hashes, pregame context, and configuration registry references. Correction replay changes current state and future forecasts without rewriting earlier issued probabilities.

For a completed game, evaluation selects the latest stored revision issued before its recorded start with matching opponents. A game discovered only after completion receives no issued probability and is excluded from live forecast metrics, even though it updates ratings. Primary live scoring requires both teams to have ten observed prior games at issuance; all-issued descriptive scores are stored separately. Evaluation joins forecasts to the latest validated outcomes.

This live policy updates after individual completions when the collector runs, whereas the historical evaluation froze daily slates. Consequently, the historical accuracy is not a measured accuracy of the live service. The first three forecasts were saved on October 13, 2026 Eastern; prospective results remain pending. Tests cover repeated polling, corrected outcomes, preservation of revisions, missed-game backfill, invalid future timestamps, failed collection, writer locking, and empty/issued/backfilled interface states. Source connectivity has been verified and automatic collection is scheduled. Public frontend hosting is deployed and verified.

11 Scheduled operation and hosting status

The GitHub Actions workflow invokes core checks and collection at minute 17 of every two UTC hours. Scheduled jobs have a three-minute timeout; manually requested full-refresh jobs allow ten minutes. Official checkout and Python setup actions are pinned to verified commit hashes. No source cache or workflow artifact is uploaded. The collector commits processed season data and status only, and refuses publication if the branch advanced during collection. An ingestion failure can publish failure status while retaining valid ratings. Scheduling is best-effort rather than an immediate final-score push service. The first hosted Ubuntu 24.04 run passed 36 core checks (nine optional frontend/fitting checks skipped), refreshed sources, and committed a valid snapshot in 40 seconds. The complete local suite passes all 45 tests. The runner platform is pinned to the verified Ubuntu release.

The workflow runs on a public repository; the account's existing zero-dollar Actions budget blocks paid usage. Allowances are shared with other repositories, so exhaustion may interrupt updates. No paid service, billing limit increase, or new credentials are configured. Cross-platform verification requires identical record structures and identities, with floating-point agreement within relative/absolute tolerance 10^{-12}; this permits platform rounding without changing the model or its stored forecasts.

The Streamlit application is deployed from the public repository's main branch, root application, and tested Python 3.12 runtime at `https://eshan-nba-elo-forecast.streamlit.app/`. Public sharing is enabled on the deployed application and the GitHub repository is public. Hosted checks verified current-season defaults, archived season switching, held-out metrics, downloadable CSV data, and light/dark appearance modes. The downloaded 2025–26 box-score table contains 2,460 rows matching every committed field; line-ending normalization does not alter the records.

An active application refreshes its saved snapshot every minute using a Streamlit fragment. This does not scrape or fit during page interactions; backend collection remains on its separate two-hour schedule. Snapshot freshness is visible, and actual future issued forecasts will establish prospective performance. Deployment verification does not imply that current season games have already produced a live accuracy estimate.

A Box-score feature derivations

All quantities below refer to one completed team-game; O denotes its opponent. Counts retain count units; rate features are fractions. Using player turnovers,

$$\widehat{P} = FGA + 0.44FTA - OREB + TOV. \tag{14}$$

The eight derived features are

$$X_1 = 100(PTS - PTS_O)/\widehat{P}, \tag{15}$$
$$X_2 = (FGM + 0.5\,3PM)/FGA, \tag{16}$$
$$X_3 = PTS/[2(FGA + 0.44FTA)], \tag{17}$$
$$X_4 = OREB/(OREB + DREB_O), \tag{18}$$
$$X_5 = DREB/(DREB + OREB_O), \tag{19}$$
$$X_6 = AST/\max(TOV, 1), \tag{20}$$
$$X_7 = TOV/(FGA + 0.44FTA + TOV), \tag{21}$$
$$X_8 = STL/\widehat{P}_O. \tag{22}$$

X_1 uses a shared-possession net-efficiency approximation. X_2 credits three-point makes with half an additional made shot; X_3 scales points by estimated shooting opportunities. Offensive and defensive rebound fractions use opposite opponent rebound categories. X_6 uses one for zero turnovers. X_7 is a turnover fraction. X_8 uses opponent estimated possessions, including the offensive-rebound subtraction. Other nonpositive denominators fail validation. All these features affect subsequent ratings, never the same game's pregame probability.

B Travel and time-zone derivation

For venue latitude φ and longitude ℓ in radians,

$$a_v = \sin^2((\varphi_2 - \varphi_1)/2) + \cos\varphi_1 \cos\varphi_2 \sin^2((\ell_2 - \ell_1)/2), \tag{23}$$
$$TD = 2R \arcsin\sqrt{\mathrm{clip}(a_v, 0, 1)}, \qquad R = 6371 \text{ km}. \tag{24}$$

Neutral-game venue cities are used rather than substituting a franchise home city. If $o(v, t)$ is a venue's dated UTC offset in hours, then $TZ = o(v_{\text{current}}, t_g) - o(v_{\text{previous}}, t_g)$. Exact arenas and actual itineraries are not measured by these proxies.

References

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